Ecosystem Services 64 (2023) 101566

Contents lists available at ScienceDirect

Ecosystem Services

journal homepage: www.elsevier.com/locate/ecoser

Full Length Article

Crowdsourcing social values data: Flickr and public participation GIS provide different perspectives of ecosystem services in a remote coastal region

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ARTICLE INFO

Keywords:
Cultural ecosystem services
Marine spatial planning
Maximum entropy models (maxent)
Nature-based tourism
Photo content analysis
Sampling bias

ABSTRACT

Spatial planning and environmental management are expected to adopt participatory processes. However, the needed spatial data on social values of ecosystem services are limited, especially for marine spatial planning for large, remote coastal areas, and the collection of such information can be time and resource intensive. Crowdsourcing techniques are cost efficient sources of social values data, but must meet the information needs of planners and managers with sufficient confidence. We evaluated the reliability of crowdsourced social values data by assessing the agreement between geotagged photos posted to the social media platform Flickr and data from an online public participation GIS (PPGIS) survey conducted to support marine spatial planning in the Kimberley region of Western Australia.

Flickr could not represent all of the social values mapped in the PPGIS dataset, was notably dominated by scenic values, and under-represented the importance of recreational fishing and Aboriginal cultural values. The Flickr dataset was more affected by accessibility from settlements and transportation infrastructure (roads, boat ramps) and more restricted spatially. Maximum entropy (maxent) distribution models developed to explore the socioecological drivers determining where social values occur also highlighted that Flickr and PPGIS samples provide different perspectives on social values, even after accounting for differences in sampling intensity. The Flickr dataset suggested that the social values occurred under narrower ranges of socioecological conditions, and the PPGIS dataset was more likely to find protected areas to be valued, likely because participants believe protected areas to support those social values, but may not themselves pursue them there.

Flickr and PPGIS crowdsourced datasets provide different perspectives on the spatial distribution of social values for several reasons: They differ in their spatial and demographic biases, as well as their measurement of revealed vs. stated preferences. Flickr is most effective at representing social values that are easily photographed and interpretable in photos, and may not capture some information needs of managers. PPGIS is under greater investigator control but, because stated preferences can be disconnected from actual use of the sites mapped, may be best suited for assessing management preferences and social acceptability. With understanding of the nuances of these datasets, crowdsourced social values data can be applied most appropriately to support successful planning outcomes.

1. Introduction

There is rapidly growing interest in using datasets generated from publicly shared social media posts to monitor and study human interactions with the environment (Barros et al., 2022; Teles da Mota and Pickering, 2020). These data are cost efficient and have large volume and high spatiotemporal detail, which has been highlighted as useful for management in a survey of local planners (Komossa et al., 2021). Social media data also represent the actual use of ecosystem services (Havinga et al., 2020; Zhang et al., 2022), avoiding the critique that many ecosystem service assessments restrict themselves to evaluating service supply or potential benefits (Mandle et al., 2021). There are legitimate concerns about the use of information shared to social media (especially their representativeness, accessibility, ethical use, and the depth and type of the information available; Barros et al., 2022; Calcagni et al., 2019; Di Minin et al., 2015; Ghermandi and Sinclair, 2019; Teles da

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https://doi.org/10.1016/j.ecoser.2023.101566
Received 20 February 2023; Received in revised form 17 September 2023; Accepted 24 September 2023
Available online 30 September 2023

Mota and Pickering, 2020; Toivonen et al., 2019; Zhang et al., 2022). Nevertheless, there is strong support that they effectively proxy patterns of visitation to natural areas over space and time (Barros et al., 2022; Ghermandi, 2022).

However, visitation rates are a coarse indicator of human interactions with the environment; much richer information can be extracted from geotagged photos publicly shared to social media. In other words, visitation rates address the 'where' and 'when' of human-environment interactions (from the framework of Di Minin et al., 2015), but social media data can also provide insight into the type of interaction ('what') and the conditions that determine 'why' it occurred at that time or place.

Content analysis of shared photos, associated text, or both allows researchers to differentiate the ecosystem services used (Clemente et al., 2019; Karasov et al., 2022; Martínez Pastur et al., 2016; Oteros-Rozas et al., 2018; Richards and Friess, 2015; van Zanten et al., 2016) or determine specific recreational activities pursued (Pickering et al., 2020). In addition to labelling the ecosystem services, content analysis may identify more detailed information about the ecosystem service providers and the quality of the service, including the species (Hausmann et al., 2018; Willemen et al., 2015) and landscape features (Fox et al., 2021, 2022; Komossa et al., 2020; Pickering et al., 2020) valued by users as providing recreational opportunities and aesthetic experiences, and the emotional responses they generate (Fox et al., 2021, 2022; Teles da Mota et al., 2022). Content analysis of social media is most commonly used for the cultural ecosystem services of recreation and aesthetic values (Zhang et al., 2022), but can support assessments of a much wider range of ecosystem services (Rossi et al., 2020; Schirpke et al., 2021; Zhang et al., 2022). Nevertheless, quantitative reviews have found content analysis to be relatively uncommon in studies of human-environment interactions using social media data (Barros et al., 2022; Ghermandi and Sinclair, 2019; Teles da Mota and Pickering, 2020; but see Calcagni et al., 2019).

Content analysis may support inferences into the drivers of ecosystem service use (Barros et al., 2022), but this can also be evaluated with statistical modelling (Toivonen et al., 2019; Zhang et al., 2022), such as using the popular modelling framework maxent (Phillips et al., 2006), to determine the spatial and temporal associations between geotagged social media posts and environmental and anthropogenic characteristics. Statistical models have been used with social media data to understand patterns of visitation to parks and natural areas (Ciesielski and Stereńczak, 2021; Donahue et al., 2018; Schirpke et al., 2021; Teles da Mota et al., 2022; Tenerelli et al., 2016), cultural ecosystem services (Clemente et al., 2019; Martínez Pastur et al., 2016; Oteros-Rozas et al., 2018; Richards and Friess, 2015; van Zanten et al., 2016), and contributions of biodiversity (Echeverri et al., 2022) and geodiversity (Fox et al., 2022) to nature-based tourism and outdoor recreation. The developed models can be applied spatially to generate maps of areas that are predicted to be suitable for supplying the ecosystem service, which is of use for spatial planning.

Although more detailed assessments of ecosystem services using social media data are burgeoning, further work is needed to demonstrate their reliability. However, while validation data exist for the coarse indicator visitation rate, evaluation data may not be available (Antoniou and Skopeliti, 2015; Barros et al., 2022) or at the appropriate scale (van Zanten et al., 2016) for detailed indicators and insights about the types of ecosystem services used and their drivers. Further, different concepts of validity may be relevant: Our usual understanding of 'validity as accuracy' – e.g., whether the georeferencing and time stamp are correct and at sufficient resolution – makes sense for validating assessments of 'where' and 'when' ecosystem services are used. In contrast, 'validity as credibility,' which is related to the data's fitness-for-use, the motivations of the contributors, and the coherence of mapped ecosystem services, may be more relevant for determining the reliability of information derived from social media posts about 'what' ecosystem services are used and 'why' (see Brown et al., 2015; Brown, 2017; Spielman, 2014).

In this context, researchers have called for social media data to be compared or integrated with data collected by traditional social science methods (Ghermandi and Sinclair, 2019; Ghermandi, 2022; Ilieva and McPhearson, 2018; Teles da Mota et al., 2022; Toivonen et al., 2019; Zhang et al., 2022). Comparative studies are rare (Calcagni et al., 2019) but growing, with comparisons of ecosystem service assessments conducted with data from social media relative to surveys (Hausmann et al., 2018; Heikinheimo et al., 2017; Moreno-Llorca et al., 2020; Wang et al., 2022), public participation GIS (PPGIS; Depietri et al., 2021; Heikinheimo et al., 2020; Levin et al., 2017; Muñoz et al., 2020; Stahl Olafsson et al., 2022), and their combinations (Komossa et al., 2020, 2021) increasingly being published.

PPGIS is a particularly useful comparator to geotagged social media data. It, too, is capable of mapping ecosystem services with high spatial detail and the collected data can be analysed in similar ways to geotagged social media data, including use of statistical modelling to determine environmental associations, interpolate mapped values across the study area, or extrapolate them to new study areas (Brown et al., 2020; Fagerholm et al., 2021; Semmens et al., 2019; Sherrouse et al., 2011). Although PPGIS is widely regarded as a valuable tool for ecosystem service assessments (Brown et al., 2012; Harrison et al., 2018), it developed in parallel to the ecosystem service concept and is often used to map social values rather than ecosystem services. Social values are assigned values; they are "the importance of places, landscapes and the resources or services they provide as defined by individual and/or group perceptions and attitudes towards a given place or landscape" (Strickland-Munro et al., 2015, p. 7). Social values are non-monetary, but span material and nonmaterial benefits people derive from the environment through goods, activities, and services, and thus are not restricted solely to cultural ecosystem services (Scholte et al., 2015). The social values included in the typology widely used by PPGIS studies correspond with cultural values for ecosystem services, especially cultural and provisioning services (Brown et al., 2020).

However, comparisons of social media data to PPGIS are complicated by the different nature of contributors' involvement (reviewed by Verplanke et al., 2016) and the information produced. First, PPGIS surveys allow participants to label sites with the ecosystem services they personally value there while, with social media data, the researcher must interpret which ecosystem services are represented in the shared content (Moreno-Llorca et al., 2020; Stahl Olafsson et al., 2022; Wang et al., 2022). Such interpretations can be challenging since the photographer's motivations to take a photo may not be apparent from the contents of the photo (Moreno-Llorca et al., 2020), the ecosystem service experienced (or desired) may not be easily photographed or interesting to share on social media (Pickering et al., 2020; Richards and Friess, 2015; Rossi et al., 2020; Zhang et al., 2022), and the text posted with photos may provide few clues about ecosystem services or social values, but instead emphasise place names (Pickering et al., 2020; Stahl Olafsson et al., 2022). PPGIS and social media may also assess different components of the ecosystem service cascade, such as service supply or use, respectively (Crouzat et al., 2022; Komossa et al., 2021). Further, social media captures revealed preferences, while PPGIS is more likely to represent stated preferences, dependent on survey design and the instructions given to participants (Crouzat et al., 2022; Komossa et al., 2020; Scholte et al., 2015; Stahl Olafsson et al., 2022). Stated and revealed preferences may differ for a variety of reasons, including accessibility limitations or recollection bias (Crouzat et al., 2022).

Comparisons of these data sources are also challenged by the different sampling procedures that generated them, with variable sampling biases, both spatially and demographically (the 'who' of Di Minin et al. (2015)'s framework). Sampling by social media has consistently been found to be constrained by accessibility, especially proximity to settled areas and roads or tracks (Zhang et al., 2022). PPGIS data is likewise influenced by accessibility, but also by the iconic landmarks labelled on the surveys and participant knowledgeability (Brown et al., 2020; Crouzat et al., 2022; van Riper et al., 2017, 2020). Each of these datasets will also be influenced by the stakeholder groups represented in their samples, as different groups value different ecosystem services and/or in different locations (Ancona et al., 2022; Dick et al., 2022; Komossa et al., 2018; Vieira et al., 2018; Wang et al., 2022). Demographic biases are a common concern for social media data, but are incompletely known given limited information about contributing users (Li et al., 2013). Demographic biases are also likely in PPGIS since voluntary participants are self-selected based on their interests and motivations about the topic (Brown, 2017; Bubalo et al., 2019; Fagerholm et al., 2021; Spielman, 2014).

Some of these differences between data sources may dissipate when comparing the results of statistical models of ecosystem services, rather than the raw data themselves. Maps of service supply may be modelled from observations of use (Karasov et al., 2022), thereby harmonising the information from different data sources. Additionally, models generated from different data sources may identify consistent socioecological drivers of an ecosystem service even if the input datasets show divergent spatial patterns. This will happen if the sites valued for an ecosystem service share similar social and environmental characteristics, despite being in different locations. For example, Donahue et al. (2018) and Komossa et al. (2020) found that models generated from social media and survey data identified relatively consistent drivers of ecosystem services. Models also give opportunities to factor out spatial patterns of sampling bias (Phillips et al., 2009), potentially increasing the comparability of models from datasets generated by different sampling procedures. Accounting for spatial sampling biases is best practice in maxent models (Merow et al., 2013), and is becoming increasingly common in studies that use them for species distribution modelling (Araújo et al., 2019). Although distribution modelling of ecosystem services is a growing application of maxent, to our knowledge, models of social values of ecosystem services have not yet accounted for the impacts of sampling bias. We believe it is important to do so in comparative evaluations of data sources, in order to ensure a fair comparison.

Further study is needed to evaluate the suitability of social media data for environmental planning, including assessments of different levels of information interpreted from the data and in a range of socioecological contexts. Social media data are perceived as most reliable in popular, accessible areas (Barros et al., 2022; Depietri et al., 2021; Heikinheimo et al., 2017; Levin et al., 2017; Stahl Olafsson et al., 2022; Teles da Mota and Pickering, 2020). However, generalisations may not be possible given the strong context dependence of cultural ecosystem services (Crouzat et al., 2022; Stahl Olafsson et al., 2022; Zhang et al., 2022). Careful evaluation of the useability of social media data in remote areas is needed, as crowdsourced data may offer the greatest gains in such regions where traditional data sources are limited.

The objective of this study was to determine whether social media data and PPGIS provide comparable perspectives on ecosystem services to support marine protected area (MPA) planning and management in the Kimberley region of Western Australia. We used content analysis of geotagged photos shared to Flickr, to draw on the high information content of shared images. Alternative social media platforms are either not primarily used for photo sharing (e.g., Twitter); have been discontinued (e.g., Panoramio); geotag only to named localities, rather than precise geographic coordinates (e.g., Instagram); or provide restrictive access to posts in a way that greatly limits their availability for research (e.g., Facebook and Instagram; Toivonen et al., 2019). Flickr is the most popular social media platform among environmental researchers (Barros et al., 2022; Ghermandi, 2022; Teles da Mota and Pickering, 2020). Flickr is also popular for users sharing nature and biodiversity photos, although it has fewer users than other social media sites (Di Minin et al., 2015; Hausmann et al., 2018; Toivonen et al., 2019). We comparatively assessed the social values identified in Flickr and PPGIS surveys to determine their levels of disagreement, and whether identified differences are mitigated or propagated by spatial generalisation or modelling. Specifically, we asked:

1. Do Flickr and PPGIS detect the same social values of ecosystem services and at similar relative abundances?
2. Do Flickr and PPGIS provide similar spatial coverage, with social values concentrated in the same locations, to similar levels?
3. Do Flickr and PPGIS identify similar associations between the social values and variables describing the region's environmental features, management regimes, and accessibility? When these associations are mapped to geographic space, are the spatial predictions of suitable locations similar?

2. Methods

2.1. Study area

The Kimberley region lies in northwest Western Australia, some 2500 km from the state capital, Perth. The study region (Fig. 1) spans 13,290 km of coastline between Port Hedland and the Northern Territory state border, encompassing over 2500 nearshore islands. The study area extends seaward, with 94 % of its total area of 875,383 km^2 in the marine environment, and nearly 40 % within the network of state and federal MPAs (with various levels of protection).

Although the Kimberley region occupies nearly one-sixth of the Australian landmass, it is sparsely populated with only 34,364 people, many living in the service and tourist towns of Broome, Derby, Kununurra and Wyndham. Nearly half of the population are of Aboriginal heritage, and some live in ~ 200 Indigenous settlements (Australian Bureau of Statistics, 2016).

The coastal zone of the study area is somewhat accessible along the sandy beaches (10 % of its length) west of Broome, and only limited land access is available along the highly crenulated rocky (22 %), muddy and mangal coasts (67 %) of the remainder of the shoreline (Short, 2006; Digital Earth Australia, 2021). Parts of the region are also seasonally inaccessible due to monsoon conditions causing flooding, isolating many smaller settlements (Bureau of Meteorology, 2021) and resulting in the low (Nov-Apr) and high (May-Sep) visitation seasons.

Despite being remote and in parts only accessible by sea, the region is vital for the Australian economy, with oil and gas, mining, agriculture, pearling, fishing and tourism being the main sectors of employment (Australian Bureau of Statistics, 2016). On average, there are 430,000 visitors to the region annually, with the Shire of Broome receiving the majority. More than half of the visitors are international tourists (pre-Covid), most commonly from Germany, USA and UK (Kimberley Development Commission, 2019; Tourism Western Australia, 2021). Visitors are drawn to the spectacular landscapes, biodiversity, marine megafauna and rich Aboriginal culture (Hughes, 2014), with heritage and history of indigenous land use dating back at least 40,000 years (e.g., Clarkson et al., 2017; Moore et al., 2020). Many visitors partake in marine cruising, beach camping, wildlife tours and expedition cruising (Beckley, 2015; Collins, 2008; Pearce et al., 2016; Scherrer et al., 2011).

In 2011, a marine parks strategy was presented for the Kimberley, allowing for an extensive network of marine parks to be established and co-managed with Traditional Owners (Government of Western Australia, 2011). Since then, several management plans for these marine parks have included social values in park planning, management and monitoring (e.g., North Kimberley or Lalang-Garram Marine Park), and an additional park was proposed for the Buccaneer Archipelago in 2019 (Department of Biodiversity, Conservation and Attractions, 2021).

Kimberley marine parks are managed mainly by the state, with co-management of the Traditional Owners who hold sea or land claims to parts of the region (e.g., Department of Parks and Wildlife, 2016). Beyond state waters, Kimberley marine parks are part of the Commonwealth North-west Marine Park Network (Director of National Parks, 2018).

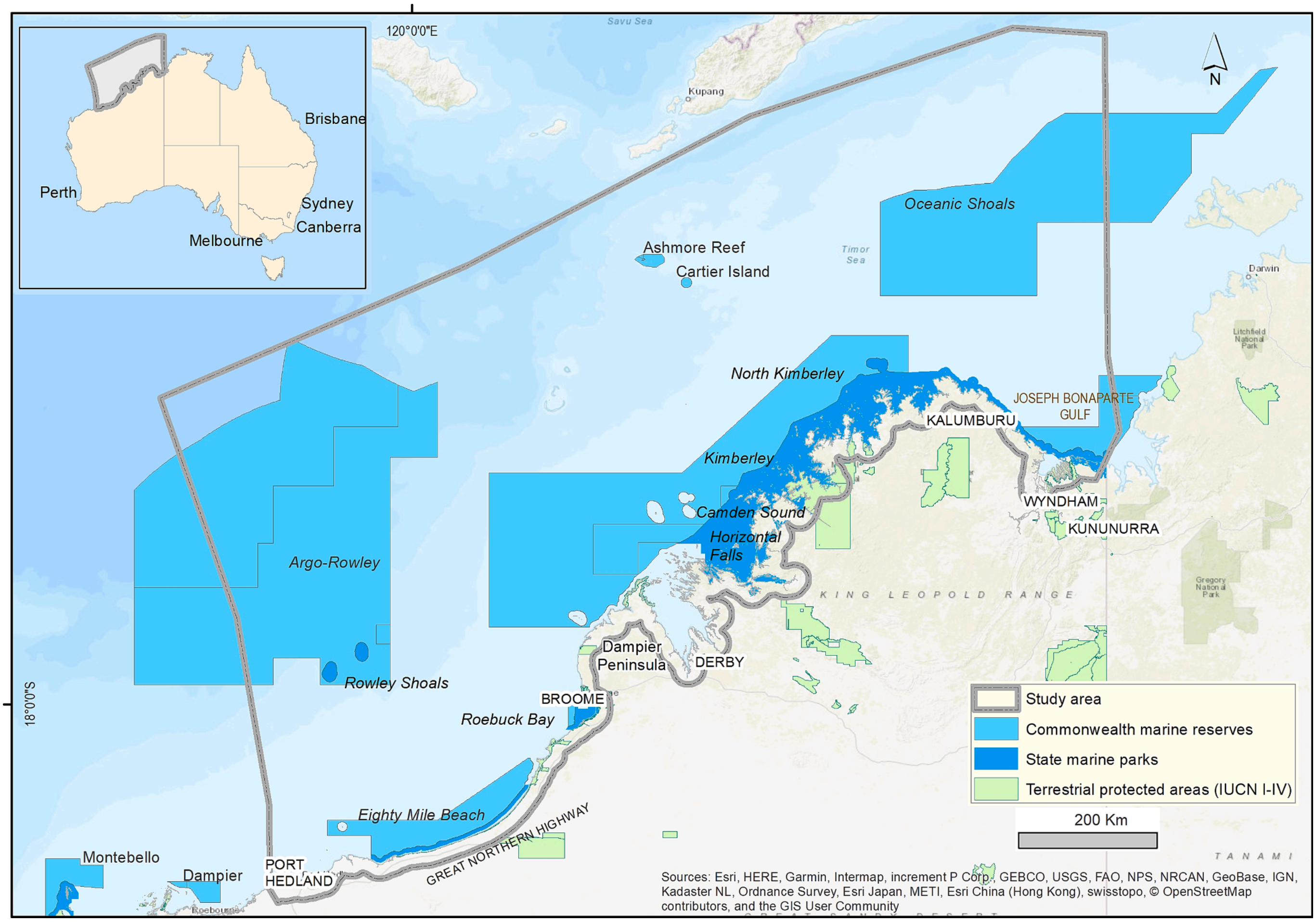

Fig. 1. Locator map of the study area, covering the coastal and marine areas of the Kimberley region of Western Australia and emphasising the size and remoteness of the region's protected areas. Locality names identify major towns and roads (black all caps text), protected areas, and natural attractions.

2.2. Social values of ecosystem services data

This study developed a social values dataset from Flickr photos within the study area and compared it to an existing PPGIS dataset. Both datasets mapped the presence of social values with point-based occurrence records, and identified the ecosystem service valued at that location.

The PPGIS dataset was collected with an online survey as part of an extensive investigation into the values people associated with existing and proposed MPAs to support marine spatial planning and management (Brown et al., 2016; Strickland-Munro et al., 2016). The social values typology used in this study was developed as part of the larger PPGIS project through an exploratory sequential mixed methods approach (Brown et al., 2017; Appendix A), resulting in a set of 14 social values mapped in the online PPGIS survey that corresponds closely to that operationalised by Brown and Reed (2000) and widely used in PPGIS studies, but reflecting region-specific social values held for the Kimberley coast, as identified in semi-structured interviews (Moore et al., 2017; Appendix A). Only the 10 social values that could be interpreted from Flickr photos were evaluated in this study (Appendix A). These values spanned provisioning and cultural ecosystem services experienced through non-consumptive use, including scenic and cultural values; direct consumptive use, e.g., recreational fishing and economic values; and indirect use, such as biological values (Table 1).

Participants in the online PPGIS survey were recruited through communications from the stakeholder organisation, local media, formal and informal networks, and direct personal contact by researchers. Participants were asked to place digital markers in a Google Maps interface identifying one of the 14 pre-determined values or 13 management preferences (Brown et al., 2016). There were no limitations to the number or proximity of markers that could be placed. Records of one 'super mapper' were thinned to 500 markers (4.4 % of the value records in the final dataset) to avoid biasing the dataset to the responses of a single, dominant participant. In total, 372 participants contributed to the online PPGIS survey, which contained 11,317 occurrence records of social values across the study area, following thinning of the super mapper's contributions.

All publicly shared geotagged photos within bounding boxes spanning the study area were identified from the Flickr Application Programming Interface (API) on 6 April 2020 (following Richards and Friess, 2015) with the following metadata recorded for each: unique user identifier; photograph ID; geographic coordinates (latitude, longitude); spatial accuracy; date- and time-stamp of the photo; user-supplied title, description, and tags; and a URL to view the photo on Flickr's website. Photos that fell outside of the study area or had insufficient spatial accuracy (i.e., accuracy levels below 15 or 16, which are the two highest levels reported by the API) were removed. There were 6,320 geotagged Flickr photos within the study area, date-stamped between June 1962 and March 2020, contributed by 406 users. Although the contributions of Flickr users were skewed (the average number of photos shared per user was 3; only 7 users shared more than 100 photos each), no individual user provided more than 5 % of photos (comparable to the super mapper's input to the PPGIS dataset following thinning), so we did not filter the Flickr dataset.

An initial ruleset to classify the 10 investigated social values (Table 1; Appendix A) was designed following studies that have

Table 1
Descriptions and examples of the 10 social values of ecosystem services mapped in an online public participation GIS (PPGIS) survey and interpreted from content analysis of Flickr photos; the abbreviation codes used in subsequent figures are also defined. Please see Appendix A for more detailed examples of each social value, and the ruleset used to classify photos to social values.

Social value	Definition	Examples	Code
Direct use, non-consumptive			
Scenic	Landscapes or attractive scenery	Gorges, waterfalls, caves, islands, beaches, sunrise/ sunset, aerial view over the landscape, moon or stars, natural processes such as fire or lightning	SCE
Aboriginal	Reflect Aboriginal history or connection with coastal and sea country	Rock art, paintings, artwork	ABC
Heritage	Reflect European history	World War Two artefacts, exploration, pastoralism	HER
Learning	Enable learning through observation or study	Clipboards, art centres, museums, photos of natural history specimens	LR
Spiritual	Sacred, religious or spiritually special places (excluding Aboriginal culture)	Churches, graveyards	SPI
Direct use, consumptive			
Recreation	Outdoor recreation activities	Off-road vehicles, bicycles, camping gear, sunbathing/ swimming, hiking, riding, diving, snorkelling, surfing, social activities such as bonfires and picnics	REC
Recreational fishing	Fishing for fish or other marine life	Fishing rods, crab pots	FISH
Economic	Natural resources which can be used by people and transportation of these resources	Oil/gas, mining, minerals, pastoralism, commercial vessels, trucks, trains	ECON
Nature-based tourism	Tourism opportunities in the natural environment	Tourism infrastructure, accommodation, cruises, cave systems, gorges, rivers, waterfalls, wildlife observation, statues or sculptures, information signs	NBT
Indirect use			
Biological	Nature appreciation, living organisms	Animals, plants, habitats such as reefs and mangroves	BIOL

performed content analysis of crowdsourced photos (Clemente et al., 2019; Muñoz et al., 2020; Richards and Friess, 2015; Tenerelli et al., 2016) and preliminary screening of photos in the database. A random sample of 100 photos was then analysed by the 3 authors independently to identify the social values represented in each photo. Any discrepancies in interpretation were discussed and used to refine the ruleset (Appendix A), which was then applied to content analysis of all photos in the database to discard photos that did not depict the natural environment or a natural resource and identify the social value(s) represented by retained photos. Photos were viewed on the Flickr website to perform content analysis and auxiliary information was drawn on, as needed, to assist interpretations, including user-supplied text uploaded with the photo and Google Maps searches to develop a contextual understanding of the area and confirm the photo location. Photos could be assigned more than one social value. An independent evaluation of the final ruleset was conducted on a random sample of 100 photos and found that interpretations of photo contents were highly consistent and repeatable: There was high agreement between coders for nearly all photos in the validation set (96 %) and total agreement for 77.6 % of the independently coded photos (please see Appendix A for detailed description of the final ruleset and the validation procedure).

Overall, 5,293 (83.8 %) of the photos analysed represented the natural environment or a natural resource. Because photos could be assigned multiple values, this yielded a sample of 7,486 occurrence records of social values.

2.3. Relative abundance of social values

The number of occurrence records for each social value was totalled to determine their relative popularity within the study area, based on either the Flickr or PPGIS dataset. The correlation in social value abundances between datasets was assessed, and major disagreements were identified with a chi-square analysis.

2.4. Spatial distribution of social values

The spatial patterns of social values were generalised using kernel density analysis, to allow intercomparison of the Flickr and PPGIS datasets. Kernel densities were generated for each of the 10 studied values (Table 1) using the geographic coordinates of the Flickr or PPGIS occurrence records and the quartic kernel function (Silverman, 1986) with a 20 km search radius and 2 km resolution. These parameters are consistent with previous hotspot mapping using the PPGIS dataset, and were selected to correspond with the size of geographic features in the study area and the coarsest mapping scale that could be used by PPGIS participants (Strickland-Munro et al., 2016). They are also coarse enough to accommodate positional uncertainties in the Flickr dataset (Zielstra and Hochmair, 2013). Kernel density analyses were conducted in ArcGIS 10.6 (ESRI; Redlands, CA, USA). Results were normalised by the abundance of the social value in a given dataset, to account for the different sample sizes of the Flickr and PPGIS datasets, and then summarised with their maximum density, skewness, and the proportion of pixels with non-zero densities, to compare broad patterns of spatial coverage and concentration provided by each dataset for each social value. Hotspots of disagreement between the Flickr and PPGIS datasets were identified for each social value by subtracting the standardised kernel density estimated from the PPGIS dataset from that produced from Flickr photos.

2.5. Socioecological drivers of social values

Maximum entropy (maxent) models (Phillips et al., 2006; Phillips and Dudík, 2008) were used to determine if the Flickr and PPGIS samples identified similar social and environmental drivers of the social values and produced similar predictions of areas suitable for the social values across the study area. Maxent is a 'presence-only' modelling technique that identifies contrasts between the environments where a social value occurs and the general range of environmental conditions available across the study area (as represented by a random background sample; Phillips et al., 2006). Maxent is among the top performing algorithms for modelling the spatial distributions of biological species (Elith et al., 2006). Maxent models make minimal assumptions (Elith et al., 2011) and are able to represent complex associations between the target and environmental predictor variables (Phillips et al., 2006; Phillips and Dudík, 2008), making it an attractive choice for modelling social values, which may integrate various uses and motivations.

Models were developed for each of the 10 social values (Table 1) using the sample of occurrence records generated from either the PPGIS survey or Flickr photos and a suite of predictor variables describing natural attractions such as beaches and waterfalls; management regimes associated with terrestrial, marine, and indigenous protected areas; and accessibility from settlements, roads, and coastal access points (Appendix B). Predictor variables were selected based on knowledge of the study area, understanding developed through content analysis of the Flickr photos, and from published models of social values in the literature. Please see Appendix B for descriptions of the expected relationships between these predictor variables and the studied social values. All variables were developed at 2 km resolution, taking care to avoid uncertainties when up-scaling from the source dataset's native resolution (Appendix B). Separate models were generated for the marine and terrestrial domains of the study area, as distinct predictor variables characterised these domains (giving a total of 40 models across all combinations of the 10 social values * 2 source datasets * 2 modelling domains). Social value occurrence records were categorised as marine or terrestrial at the native 1:100,000 resolution of the coastline dataset and a 1.5 pixel-width buffer of the coastline was included in both terrestrial and marine models to ensure coastal records were correctly handled at the model resolution.

We also factored out sampling bias in the models (Phillips and Dudík, 2008; Phillips et al., 2009), enabling us to determine if Flickr and PPGIS samples captured similar underlying socioecological drivers of the occurrence of social values despite superficial differences due to the varying influence of accessibility on each sampling approach. Bias layers were developed with maxent models of sampling intensity using the full set of all occurrence records in either the PPGIS or Flickr dataset and a subset of predictor variables describing broad patterns of accessibility (terrestrial bias: urban areas, state roads, road density, and coastal access; marine bias: urban areas, coastal access, and mainland distance; Appendix B). Separate bias layers were generated for the Flickr and PPGIS samples. They modelled the overall spatial distribution of sampling well, and are described more thoroughly in Appendix C.

Models were generated for social values with $\geq$ 10 occurrence records at the 2 km analysis resolution. All models were generated with a random subsample of 75 % of the occurrence records, retaining 25 % of records for testing. This was replicated 10x and averaged. Maxent's default settings were used for the background sample, and for feature types and regularisation, which influence the complexity of the model (Phillips et al., 2006; Phillips and Dudík, 2008). Maps of predicted suitability were produced and model performance was evaluated for both the continuous maxent outputs and for binary maps of suitable (or highly suitable) vs. unsuitable areas for each social value (please see Appendix D for information about how the suitable and highly suitable thresholds were set). The continuous maxent predictions of suitability were evaluated using the area under the receiver operating characteristic (ROC) curve (AUC), which is a thresholdless evaluation metric synthesizing estimates of model sensitivity (i.e., detection rate, or the proportion of occurrence records contained within the predicted suitable area) and specificity (related to the false positive rate) across all possible thresholds. AUC scores greater than 0.7 generally indicate a successful model (Franklin, 2010). The binary suitability classes were evaluated with the detection rate and with the area they identified to be suitable for the social value. (Please note that AUC scores were assessed on the withheld set of test records for each replicate model and averaged, while the suitability classes were produced by thresholding the overall estimates of suitability averaged across all replicates and their detection rates were assessed against all occurrence records of the social value.)

The following estimates of variable usage were used to derive inferences about the socioecological drivers of social values: (1) Variable importance was estimated as the reduction in model performance when a given variable was randomly permuted. (2) The associations between social values and predictor variables were illustrated with marginal response curves, which plot the relationship between predicted suitability for a social value and a given variable when holding all other variables at their averages.

Models generated from Flickr and PPGIS datasets were compared by (1) reciprocally evaluating the performance of the Flickr model against the PPGIS dataset and vice versa, using the AUC and detection rates (as above), (2) comparing spatial overlap between the suitable and highly suitable areas identified by each model, and (3) inspecting the variable usage diagnostics for similarities and differences. Spatial overlap was estimated with Sorenson's index of similarity, which is calculated as: $(2 * A_{overlap}) / (A_{Flickr} + A_{PPGIS})$, where $A_{overlap}$ is the area predicted to be suitable (or highly suitable) by both models and A_{Flickr} and A_{PPGIS} are the areas predicted to be suitable (or highly suitable) by the models produced from the Flickr and PPGIS datasets, respectively.

3. Results

3.1. Relative abundance of social values

There was a moderate correlation between the abundances of social values between datasets (r = 0.625; Fig. 2), which declined slightly when evaluated separately for terrestrial (r = 0.549) or marine records (r = 0.546). Biological and scenic values were abundant in both datasets; learning, heritage, economic, and spiritual values were uncommon (Fig. 2).

However, the social values made up significantly different proportions of the PPGIS and Flickr datasets ($\chi^2 = 2762$, df = 9, $p < 0.05$); only biological, recreation, and spiritual values had consistent frequencies between the datasets (Fig. 2). The largest proportional deviations were for recreational fishing and Aboriginal values, which were only 6–17 % as frequent in Flickr photos than in the PPGIS survey, and economic and scenic values, which were more than twice as frequent in Flickr than in PPGIS, relative to the total sizes of these samples (Fig. 2).

3.2. Spatial distribution of social values

Flickr and PPGIS samples showed notably different spatial distributions. PPGIS provided greater coverage of the study area. When averaging over social values, 14 % of the study area was sampled by the PPGIS dataset (i.e., had non-zero kernel density estimates); this was only

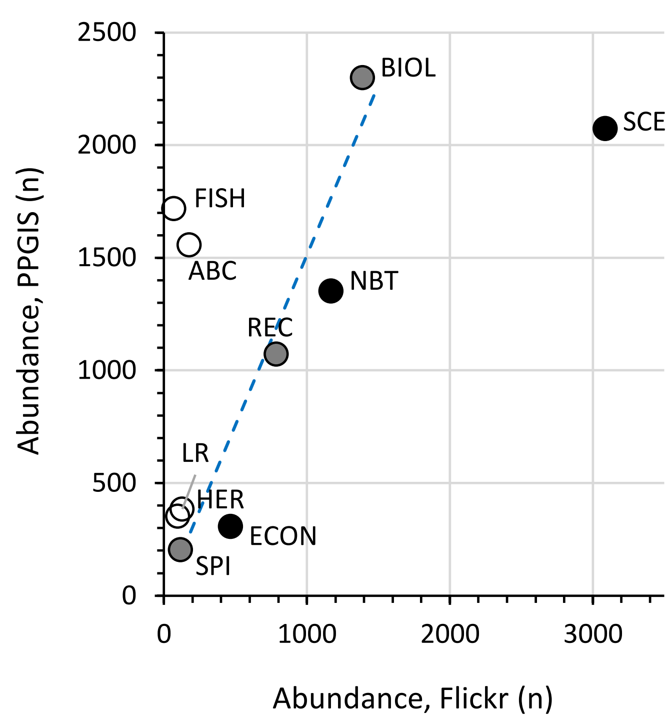

Fig. 2. Relationship between the abundance of social values of ecosystem services, as sampled by Flickr and PPGIS datasets in the Kimberley region of Western Australia. The dashed line indicates where social values would plot in this graph if the relative abundance of a social value was consistent in both datasets. Points are colour-coded by departure from these expectations: grey points indicate social values that were within 10% of expectations; white and black points indicate social values for which the Flickr dataset had large negative and large positive deviations from expectations, respectively. Social value abbreviations are defined in Table 1.

Table 2
Summary statistics of the spatial kernel density layers produced from Flickr or PPGIS samples of the 10 studied social values for the study area overall, as well as the terrestrial and marine portions of the study area. Kernel density results are summarised with the proportion of nonzero pixels, indicating the overall spatial coverage of the sample; the maximum estimated density, indicating the highest concentration of occurrence records of a social value anywhere in the analysis area; and the skewness, which is inversely related to how evenly the mapped social value records are distributed spatially. Results presented are the averages across the 10 values.

	Flickr			PPGIS		
	Overall	Terrestrial	Marine	Overall	Terrestrial	Marine
Proportion nonzero	0.04	0.35	0.03	0.14	0.85	0.11
Max	0.73	0.73	0.73	0.21	0.21	0.21
Skewness	30.38	10.87	34.81	14.53	5.62	16.71

4 % for the Flickr sample (Table 2). The Flickr dataset also showed greater concentration of sampling, with consistently higher maxima and greater skewness of the kernel density estimates (Table 2). Both datasets provided better coverage and less skewed sampling distribution of the terrestrial than the marine area (Table 2).

These patterns are also evident in the kernel density difference layers, highlighting relative hotspots of sampling from each dataset (Fig. 3; Appendix E). Greater concentration of sampling by Flickr was especially prominent near the few major towns (Port Hedland, Broome, and Derby). The PPGIS sample showed the entire coastline to be valued: sampling hotspots of PPGIS relative to Flickr were well distributed throughout the terrestrial and coastal region, including in the remote northern Kimberley where Flickr photos were sparse.

3.3. Socioecological drivers of social values

Maxent models of the social values (Appendix F) performed well, with AUC > 0.7 for all models except for terrestrial learning values (AUC = 0.671) and marine economic values (AUC = 0.678), both based on the Flickr sample (Table 3). Overall, model performance was slightly higher for models developed from Flickr occurrence records and for the marine portion of the study area (on average, $AUC_{Flickr,terr}$ = 0.850, $AUC_{Flickr,mar}$ = 0.890, $AUC_{PPGIS,terr}$ = 0.776, $AUC_{PPGIS,mar}$ = 0.925). Samples were too small to produce marine models from the Flickr dataset for Aboriginal, heritage, learning, and spiritual values.

The PPGIS-based models predicted gradual gradients in suitability for the social values. As a result, relatively large areas were predicted to be suitable in the thresholded results (36.2 % of the terrestrial area and 9.2 % of the marine area were identified as suitable when averaging across all social values) and there was a notable decline in the area found to be highly suitable (on average, 6.6 % and 2.9 % of the terrestrial and marine areas, respectively; Fig. 4, Appendix F) and in the detection rates between the suitable (90 % detection rate, by definition) and highly suitable classes (~60 % detection rate for both terrestrial and marine occurrence records, on average; Fig. 4), since a number of occurrence records received predictions of moderate but not high suitability. In

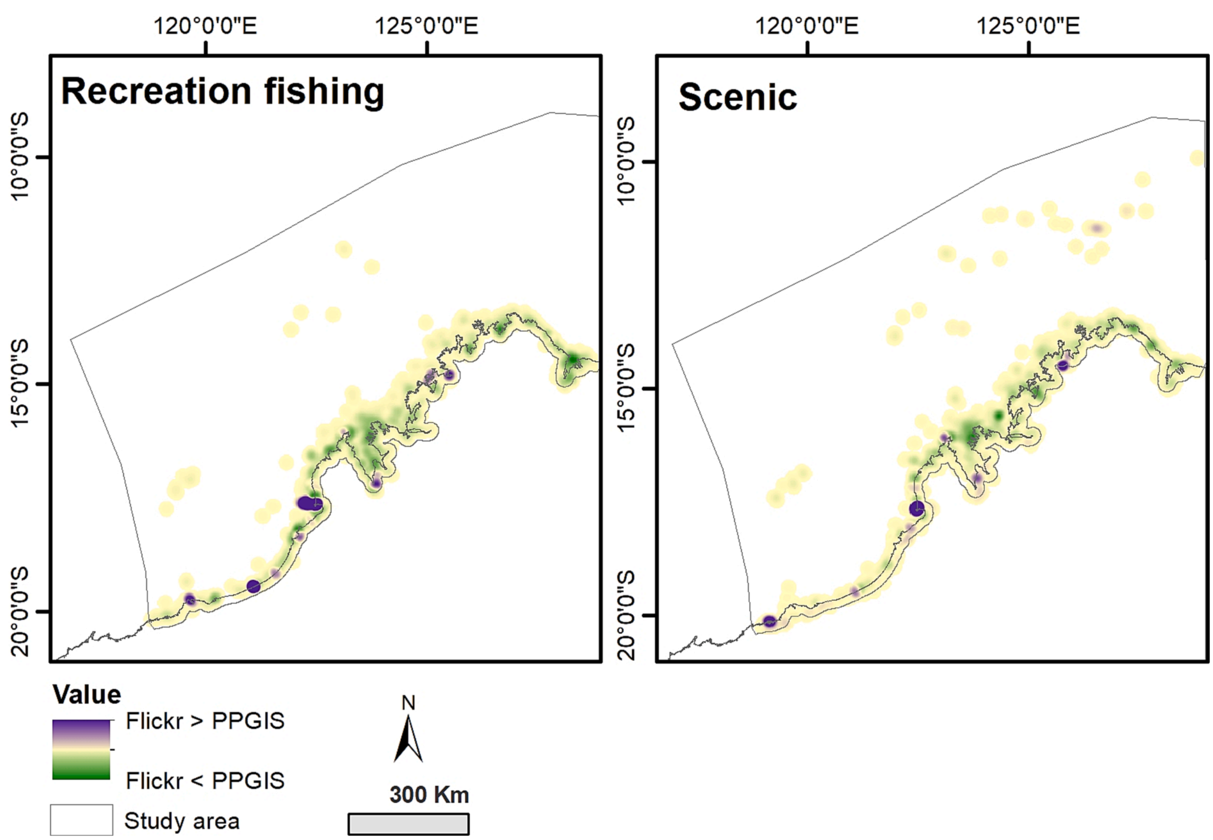

Fig. 3. Kernel density difference maps highlighting hotspots of large differences in the abundance of occurrence records between Flickr and PPGIS samples for illustrative social values: recreational fishing (left) and scenic values (right). The colour stretch in the provided maps is centred at difference values of 0 (tan colours), with purple colours indicating a notably higher concentration of occurrence records in the Flickr than the PPGIS dataset and green colours showing the reverse. The background colour of white is used for areas that were not sampled by either dataset. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Estimates of maxent model performance and consistency at predicting the spatial distribution of the 10 studied social values of ecosystem services based on socio-ecological predictor variables and occurrence records of the social values as sampled by either Flickr or PPGIS datasets. Separate models were constructed for marine and terrestrial portions of the study area for all social values with sufficient sample size ('–' indicates a model could not be developed). Model performance is estimated with the area under the receiver operating characteristic curve (AUC); AUC > 0.5 indicates a model performs better than chance, and AUC > 0.7 indicates a 'useful' model. The consistency between Flickr- and PPGIS-based models is measured by reciprocally evaluating the ability of models trained on the Flickr dataset to predict the occurrence of social values observed in the PPGIS dataset, and vice versa.

	Terrestrial				Marine			
Training dataset:	Flickr		PPGIS		Flickr		PPGIS	
Evaluation dataset:	Flickr	PPGIS	PPGIS	Flickr	Flickr	PPGIS	PPGIS	Flickr
Social value								
Aboriginal	0.864	0.616	0.735	0.805	–	–	0.956	–
Economic	0.956	0.751	0.788	0.932	0.678	0.605	0.877	0.682
Heritage	0.908	0.722	0.843	0.978	–	–	0.928	–
Recreational fishing	0.885	0.655	0.825	0.939	0.892	0.574	0.929	0.762
Learning	0.671	0.710	0.735	0.925	–	–	0.917	–
Biological	0.853	0.612	0.755	0.704	0.970	0.697	0.901	0.819
Nature-based tourism	0.859	0.737	0.786	0.843	0.961	0.716	0.909	0.837
Recreation	0.866	0.731	0.793	0.821	0.934	0.677	0.932	0.848
Scenic	0.819	0.710	0.773	0.732	0.903	0.802	0.922	0.774
Spiritual	0.814	0.572	0.731	0.847	–	–	0.974	–

contrast, the Flickr-based models identified strong contrasts in suitability, with extremely high levels of suitability predicted at most occurrence records. This caused small areas to be identified as suitable by the Flickr-based models (11 % of both terrestrial and marine areas, on average, but note the marine estimate is influenced by the large outlier result of economic values; Fig. 4, Appendix F) and maintained high detection rates for the highly suitable class (80 % terrestrial, 75 % marine; Fig. 4).

These differences in the total area predicted to be suitable drove large disagreements in the mapped results between models developed from Flickr and PPGIS datasets, with Sorenson's similarity indices typically ranging between 0.2 and 0.3 (Fig. 4). However, the more extensive predictions from the PPGIS-based models were surprisingly successful at predicting the occurrences of social values in the Flickr dataset. In the terrestrial portion of the study area, the models generated from PPGIS occurrence records even showed *improved* performance when evaluated on the Flickr dataset, rather than on the PPGIS sample of social values, with average AUC on the Flickr sample of AUC = 0.853 (Table 3) and higher detection rates for the high suitability class (Fig. 4). On average, for the terrestrial portion of the study area, 73 % of the area identified to be suitable by the Flickr-based models was also predicted to be suitable by the counterpart models produced from the PPGIS sample (Fig. 4). These similarities weakened for the marine portion of the study area and the highly suitable class, and when evaluated in the opposite direction, with Flickr-based models typically performing poorly when evaluated on the PPGIS sample (Fig. 4, Table 3).

Spatial agreement in the areas predicted to be highly suitable was modest (Fig. 4) and largely constrained to the main settlements for all social values, and to prominent landmarks such as tourist sites with aboriginal rock art (Aboriginal values) and famous waterfalls (nature-based tourism and scenic values), as well as actively mined islands (economic and recreation values). However, there were more extensive areas of agreement in the nearshore areas for marine models of biological and scenic values (Fig. 5).

Models developed from Flickr and PPGIS datasets generally disagreed about which predictor variables were most important for explaining the distributions of social values, even in cases where consistencies between the identified suitable and highly suitable areas were noted. Correlations in variable importance scores were not strong between models developed from each dataset (terrestrial models: r_{avg} = 0.36, range = [0.04, 0.53]; marine: r_{avg} = 0.40, range = [0.17, 0.77]). In some cases, divergent associations were found between a social value and a given predictor variable (Appendix F). There was no consistent tendency for the cumulative contributions of environmental and access variables to differ between terrestrial models based on either dataset, but access variables had stronger contributions to marine models developed from Flickr than to counterpart models from PPGIS occurrence records (Appendix F). In both domains, management variables were more important to models developed from the PPGIS dataset, which found most social values to be more likely within or adjacent to protected areas, as seen in response curves (Appendix F) and in the maps of predicted suitability generated by the models (Fig. 5, Appendix F). Similar patterns were not evident in the model results developed from the Flickr dataset.

4. Discussion

Geotagged photos shared to Flickr yielded a reasonable volume of data about social values of ecosystem services used in the natural coastal environments of the Kimberley region of Western Australia, an encouraging outcome given the remoteness of the region. However, our evaluations highlighted discrepancies between the Flickr sample and a published PPGIS survey (Brown et al., 2016; Strickland-Munro et al., 2016) of social values across the region. While maxent models of the spatial and socioecological distributions of social values helped to understand the differences between these datasets, they did not mitigate these inconsistencies. Thus, Flickr and PPGIS samples differed along all evaluated dimensions of Di Minin et al. (2015)'s framework ('what,' 'where,' 'why', and, by inference, 'who').

4.1. Flickr is not equally amenable to all social values

Despite broad similarities between the popularity of social values in the Kimberley when sampled with Flickr or PPGIS, there were important discrepancies in the values assessed. A major limitation of the Flickr sample was that some social values could not be interpreted from photo content analysis and were excluded from study. This included 4 values within the PPGIS dataset – life sustaining, therapeutic/health, intrinsic/existence and wilderness/pristine values (Brown et al., 2016), and values identified in an interview-based PPGIS study in the region but not included in either dataset studied here – social interaction and memories, experiential, subsistence, and bequest values (Moore et al., 2017). In contrast, other researchers have found more diverse recreational uses of natural areas in social media, since they were not limited to an *a priori* set of activities included by survey designers (Heikinheimo et al., 2017).

Nevertheless, 10 values were represented in both datasets, but in sometimes strikingly different amounts. We observed significant discrepancies in the relative abundance of over two-thirds of the studied

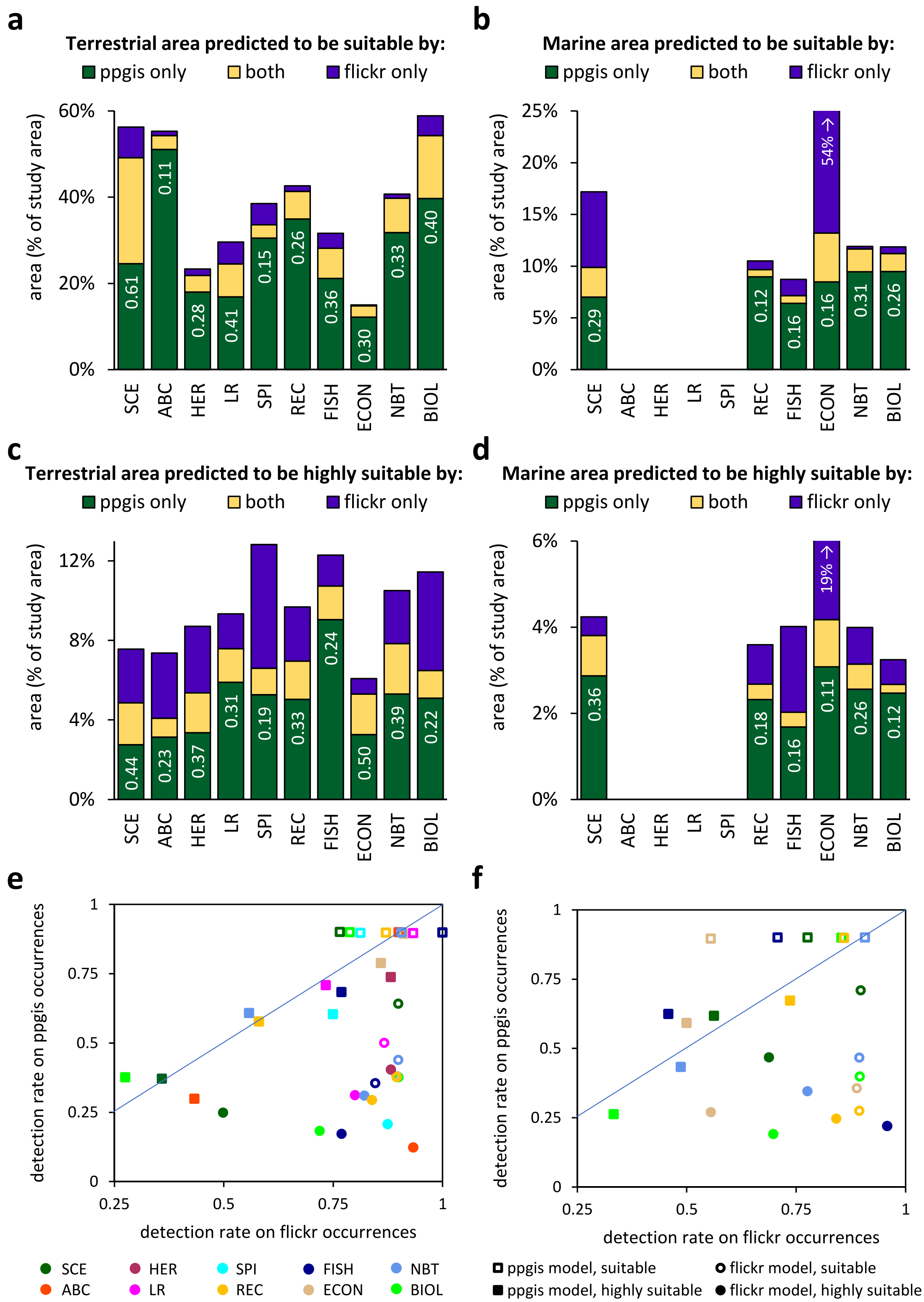

Fig. 4. Comparisons of the categorical results of the maxent models of social values that have been thresholded into suitability classes. Panels (a) through (d) show the area predicted to be (a, b) suitable and (c, d) highly suitable for each social value by models developed on Flickr and PPGIS samples, as well as the amount of spatial overlap between counterpart models for the (a, c) terrestrial and (b, d) marine portions of the study area. Bars are labelled with the Sorenson index of similarity comparing the results of the Flickr- and PPGIS-based models. Please note that, for clarity, the y-axis range differs between panels, and has been truncated for the outlier (economic values) in panels (b, d). Panels (e) and (f) show the detection rates of the suitable and highly suitable classes predicted by the (e) terrestrial and (f) marine models when reciprocally evaluated on each dataset. Points falling to the right of the 1:1 line (blue) indicate stronger performance (a higher detection rate) when evaluated on the Flickr than the PPGIS sample. Social value abbreviations are defined in Table 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

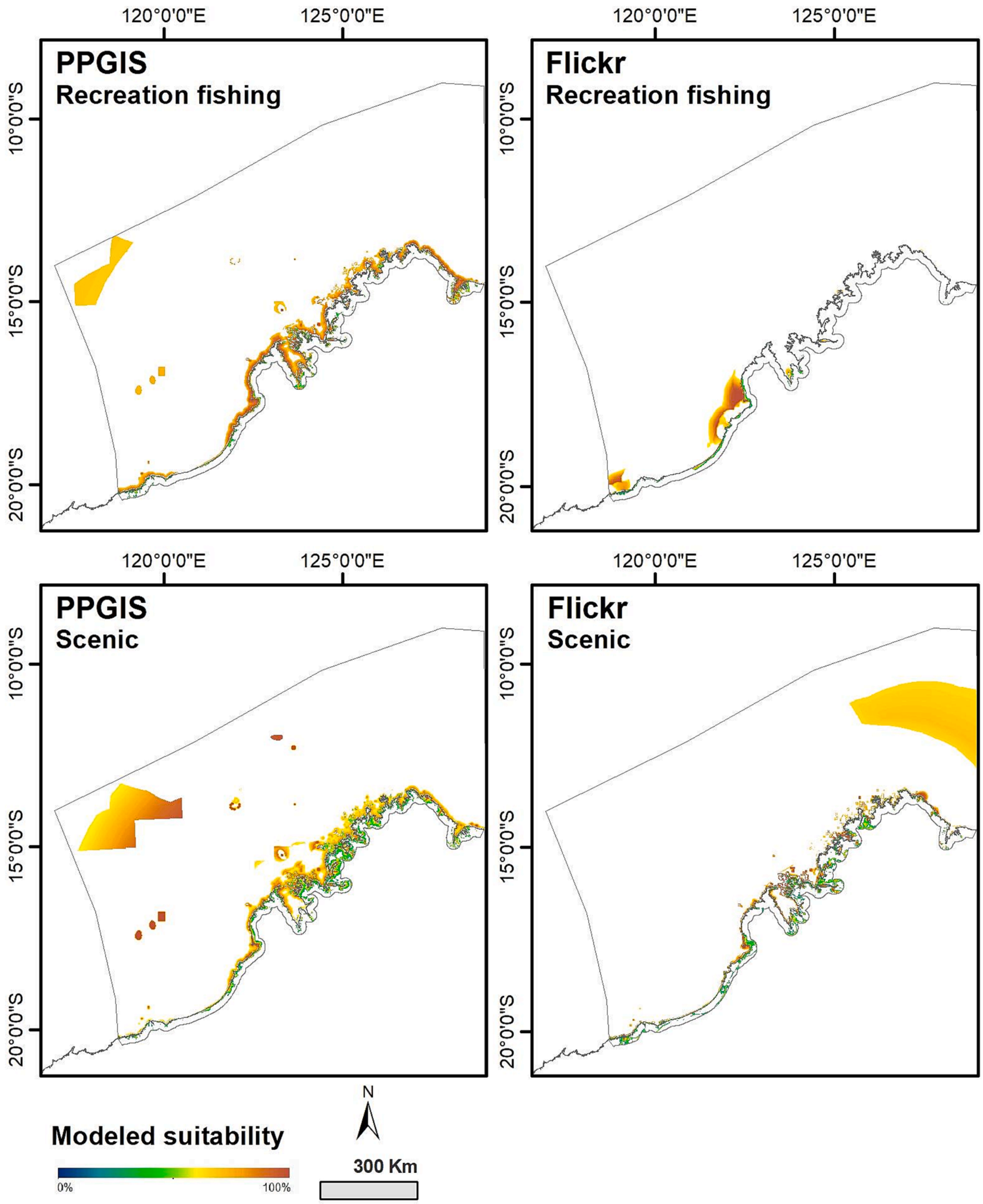

Fig. 5. Maps of the area predicted to be suitable by maxent models of the distributions of illustrative social values: recreational fishing (top) and scenic values (bottom), developed from both PPGIS (left) and Flickr (right) samples. Suitable areas are mapped in shades of colour corresponding to the modeled suitability estimate at each pixel. These suitability estimates are the continuous logistic output of maxent, which are proportional to the probability a social value will be present at a given location, based on its socioecological characteristics. Unsuitable areas (and areas outside of the study area) are depicted in white. Results from both terrestial and marine models are mapped in the same panel. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

social values. Flickr over-represented scenic values of aesthetically pleasing landscapes, consistent with previous evaluations of Flickr alone (Clemente et al., 2019; Figueroa-Alfaro and Tang, 2017; Martínez Pastur et al., 2016; Retka et al., 2019) and in comparisons with PPGIS (Depietri et al., 2021; Muñoz et al., 2020). This is likely because of the ease and motivations of photographers of capturing aesthetic values with photography (Figueroa-Alfaro and Tang, 2017; Yoshimura and Hiura, 2017).

Intangible values, such as spiritual and learning values, were uncommon in Flickr photos, as such values are difficult to photograph or to interpret in content analysis (Richards and Friess, 2015; Zhang et al., 2022). However, these values were similarly uncommon in the PPGIS dataset. Although we might expect intangible values to be better captured by PPGIS surveys than from photo content analysis, Brown et al. (2017) suggest that they are also uncommonly mapped in PPGIS because they are cognitively challenging to participants, which is known to influence the quantity and quality of PPGIS data (Fagerholm et al., 2021).

The most under-represented social values in Flickr photos, relative to the PPGIS dataset, were Aboriginal cultural values and recreational fishing, both of which are known to have high prominence in the region (Beckley, 2015; Hughes, 2014). For example, aerial surveys of coastal use found recreational fishing to be the most common activity of visitors (Beckley, 2015), which is more in line with the results of the PPGIS survey than the contents of photos posted to Flickr.

4.2. The Flickr sample is likely to be biased demographically

Some of these discrepancies in the relative abundance of social values between the datasets may be related to demographic differences between Flickr and PPGIS participants. The composition of social media users is unknown, but is expected to be unrepresentative of the general population (Li et al., 2013). Moreover, different social media platforms have different user bases demographically (Heikinheimo et al., 2017) and among different nationalities (Ghermandi and Sinclair, 2019; van Zanten et al., 2016), and may represent different social values (Echeverri et al., 2022; Havinga et al., 2020) as their users have different motivations to post (Oeldorf-Hirsch and Sundar, 2016) and may share different types of content (Di Minin et al., 2015; Ghermandi and Sinclair, 2019; Teles da Mota and Pickering, 2020). Thus, there are recommendations to combine data from multiple social media platforms to better represent the community and their values (Ghermandi, 2022; Ghermandi and Sinclair, 2019; Heikinheimo et al., 2020). Nevertheless, comparative studies have found that the results from alternative social media platforms are more strongly related to each other than they are to data generated by different means, such as PPGIS (Donahue et al., 2018; Heikinheimo et al., 2020). Therefore, although our study evaluated only a single social media platform, we believe the results of our comparisons of Flickr to PPGIS surveys of social values are likely to be representative of the relative performance of other platforms that support photo and text-based sharing.

Surveys have variously found Flickr users to be unbiased by age (Hausmann et al., 2018), or biased to older age groups (Heikinheimo et al., 2017), especially men and professionals (Lenormand et al., 2018), which is broadly consistent with the sample of PPGIS participants in this study, who tended to be older and well-educated (reported in Brown et al., 2016). However, more detailed comparisons between the PPGIS and Flickr contributors are not possible since we did not survey the Flickr users who shared photos of the Kimberley region.

The motivations of the PPGIS and Flickr contributors likely differed, leading to differences in the relative abundances of values. For example, pro-conservation preferences dominated among the PPGIS participants (Strickland-Munro et al., 2016), which may be why economic values were mapped much less frequently by the PPGIS survey than by content analysis of Flickr photos. The PPGIS respondents knowingly provided input for the purpose of MPA planning and may have chosen to participate because they believe MPAs are important. In contrast, Flickr users share photos to document their lives and connect with a community (Oeldorf-Hirsch and Sundar, 2016) and interpretations of photo contents may inadequately capture the photographer's motivations for taking and sharing the picture (Moreno-Llorca et al., 2020). Thus, photos of natural resource infrastructure may not equate to preferences for economic development.

4.3. Accessibility biases in the Flickr sample

Although the Flickr and PPGIS datasets provided reasonably similar sample sizes, the PPGIS sample was much better distributed spatially. The Flickr sample was more restricted by accessibility and concentrated around the main settlements, a pattern which is commonly reported (Zhang et al., 2022). Although others have interpreted this general pattern to suggest that social media data will be most reliable in regions with high populations and accessibility (Heikinheimo et al., 2017; Levin et al., 2017; Stahl Olafsson et al., 2022; Teles da Mota and Pickering, 2020), we found that geotagged Flickr photos do provide useful information in a remote, sparsely populated region, though the associations with accessibility are maintained within the region.

Not only were the social values in the Flickr sample restricted spatially, they also occurred under a narrower range of social and environmental conditions, as represented by the predictor variables provided to the maxent models, and tended to be nested within the more generalised distributions indicated by the PPGIS sample. This is seen in the lower amounts of suitable and highly suitable area predicted by the maxent models based on Flickr relative to PPGIS occurrence records, the asymmetric contribution the area of agreement made to the total area predicted to be suitable by Flickr- and PPGIS-based models, and also by the fact that models developed from the Flickr dataset, in particular, tended to find strong contrasts between highly suitable areas containing most of the occurrence records and unsuitable areas. Because the Flickr sample of social values exhibited greater specialisation to narrow environmental requirements, the models it supported had higher apparent performance when evaluated with the AUC, which is consistent with findings in the ecological literature that distribution models perform better for specialist rather than generalist species (Evangelista et al., 2008; Tsoar et al., 2007).

Much of this specialisation in the Flickr sample of social values, especially in the marine area, was associated with accessibility – model response curves often revealed greater likelihood of social values occurring closer to the mainland (in marine models), urban areas, settlements, and main roads, and in areas with higher road density (Appendix F). The importance of accessibility and infrastructure to patterns of actual service use, as captured by the Flickr sample (Levin et al., 2017; Li et al., 2013; Retka et al., 2019), is commonly observed (Clemente et al., 2019; Muñoz et al., 2020; Oteros-Rozas et al., 2018; Richards and Friess, 2015; Westcott and Andrew, 2015). Our maxent models detected associations with accessibility despite use of bias layers to factor out patterns of sampling intensity. The bias layers represented impacts of accessibility on the pooled sample of social values; thus models detected patterns between individual social values and accessibility that differed from the sample as a whole.

Our use of a bias layer to account for variation in sampling intensity is novel for spatial assessments of cultural ecosystem services. We believe it is important to do so, as in models of biological species (Merow et al., 2013), but suggest that it depends on whether the focus of the model is the supply, demand, or use (Tallis and Polasky, 2011) of the service. Supply of cultural ecosystem services is determined by the biophysical and social environment, while demand is a function of the population of beneficiaries and accessibility of the service (Andrew et al., 2015; Tallis and Polasky, 2011); use integrates these components (Tallis and Polasky, 2011). If the goal is to understand and map ecosystem service use, accessibility is an important contributor (e.g., Westcott and Andrew, 2015), not a source of bias. However, strong accessibility biases may mask the underlying environmental drivers of services (Andrew et al., 2015). Biases also complicate comparisons of the Flickr and PPGIS samples, which followed very different sampling distributions and may assess different components of the ecosystem service cascade (use vs. supply, respectively). A previous modelling study similarly found accessibility to be a stronger driver for Flickr photos than for a PPGIS dataset (Muñoz et al., 2020), but uncontrolled differences in sampling effects may preclude the ability to identify more interesting similarities (or differences). Thus, we included estimates of the spatial biases of Flickr and PPGIS samples (Appendix C) in our models to support more controlled comparisons. Nevertheless, even after accounting for sampling effects, we found considerable differences between the model predictions generated from Flickr and PPGIS samples, suggesting that spatial sampling biases are not solely responsible for the differences between these datasets.

4.4. Flickr and PPGIS identify different socioecological drivers for social values

Although the social values sampled by Flickr tended to occur in a subset of sites where they were found to be supported in the PPGIS dataset, and models generated from the PPGIS sample were able to effectively predict the social values mapped by geotagged Flickr photos, this was not because the maxent models identified underlying similarities in the socioecological drivers of the social values. Variable usage diagnostics highlighted that the distribution models developed from Flickr and PPGIS datasets drew on different variables (Appendix F). Even in situations where prominent landmarks, such as waterfalls, were identified as highly suitable by both models, this was not caused by a shared reliance on a closely associated predictor, such as the distance to waterfalls variable (Appendix B, F).

The inconsistencies between the Flickr- and PPGIS-based models are partly due to the heterogeneous nature of the social values, which do not neatly capture single ways to use the environment. For example, biological value encompasses exposure to biodiversity in general, appreciation of particular species or habitats, and encounters with captive biodiversity in wildlife parks or botanical gardens, each with disparate socioecological associations. The complex nature of the social values is evident in the maxent models, which drew on many variables with relatively low importance scores to synthesise the various associations of the different subtypes of each social value.

The Flickr and PPGIS samples may emphasise different aspects of the studied social values that are pursued by different stakeholder groups, leading to discrepancies in the model results. For example, recreational fishing can be pursued in the ocean or in inland waterbodies and estuaries. There was a clear dichotomy between the terrestrial Flickr and PPGIS samples, which primarily represented fishing from the beach and inland fishing, respectively, as evidenced by the mapped results (Fig. 5) and divergent associations with the coast distance and inland water variables (Appendix F). Such inconsistencies are likely due to different demographic biases between the Flickr and PPGIS contributors, which can propagate into different perspectives of the locations and socioecological conditions in which social values are pursued, in addition to differences in the relative popularity of social values discussed earlier. Flickr's sample of recreational fishers in our study area appears to be small and unrepresentative of the PPGIS participants. In an earlier analysis of the PPGIS survey used in our study, Munro et al. (2017) found that mapped recreational fishing values differed strikingly between residents and visitors to the region. Based on these spatial patterns, our Flickr sample appears most consistent with the recreational fishing values of visitors. Likewise, previous studies have found social media datasets to correspond with the preferences of visitors and to be concentrated to the sites most popular among tourists (Depietri et al., 2021; García-Palomares et al., 2015).

Recreational fishing was not the only social value to show opposing socioecological associations when modelled from Flickr vs. PPGIS samples. The most notable contradictions occurred with management variables representing the proximity of various categories of protected areas, likely due to differences between the revealed and stated preferences sampled by Flickr and PPGIS, respectively, and effects of the knowledgeability of the PPGIS contributors. Our maxent results suggest that PPGIS participants value protected areas (similar to the findings of van Riper et al., 2017) for diverse direct and indirect uses (Appendix F), perhaps because they believe reserves protect valuable sites, or they trust the management of protected areas is conducive to these values. In addition, less knowledgeable PPGIS participants may be influenced by the landmarks labelled on the base maps provided (Crouzat et al., 2022; van Riper et al., 2020), which likely explains the influence we observed of protected area boundaries, which were included in the mapping portal (Moore et al., 2017). The Flickr sample did not reveal that people use these ecosystem services in protected areas, instead tending to identify social values outside of protected areas (Appendix F), potentially due to restrictions on the activities that can be pursued in protected areas.

5. Conclusions and recommendations for managers

Crowdsourced information from social media platforms or online PPGIS mapping tools are an attractive, efficient means to gather data about use and social values of ecosystem services. Crowdsourced methods may be especially beneficial in large, remote areas such as the Kimberley region of Western Australia, which challenge rigorous on-the-ground sampling. However, our comparative study shows that these sampling approaches have multi-faceted influences on the data collected and we conclude that data from social media and PPGIS surveys are incommensurate sources of information about social values for ecosystem services. While it is encouraging that the areas identified as valued by the Flickr sample tended to be nested within the broader area valued by PPGIS participants, the differences between samples could not be mitigated by higher-order analyses such as maxent models accounting for sampling biases. Thus, while social media data provide valuable information about human-environment interactions, including in the remote Kimberley region of Western Australia, they offer a different perspective than does PPGIS. This includes differences in the relative popularity of social values, the spatial pattern of locations valued, and the inferred socioecological drivers of social values in the study area. Our conclusions agree with those made by previous comparative assessments of social media and PPGIS surveys (Depietri et al., 2021; Heikinheimo et al., 2020; Levin et al., 2017; Moreno-Llorca et al., 2020; Muñoz et al., 2020; Stahl Olafsson et al., 2022), suggesting they are likely to be general. In this section, we synthesise our conclusions with insights from previous research to identify emerging generalities and make recommendations for the use of Flickr and PPGIS to represent social values for environmental management and planning.

Flickr allows managers to assess patterns of actual visitation in considerable spatiotemporal detail (Wood et al., 2013). Although not explored here, photo timestamps can be used to monitor changing usage patterns (Heikinheimo et al., 2017; Retka et al., 2019; Tenkanen et al., 2017; Walden-Schreiner et al., 2018), including responses to management actions (Volenec et al., 2021). Flickr photo analysis can also be used as a preliminary investigation into the breadth of social values and their geographic footprints and hotspots (Depietri et al., 2021; Heikinheimo et al., 2017), or can provide early detection of novel uses or newly valued locations, informing subsequent participatory mapping. However, Flickr datasets are uncontrolled by the investigator and may be disconnected from the information needs of the managers regarding particular social values or locations in their planning area. As we observed in the present study, Flickr may misleadingly represent the relative popularity of social values, giving scenic values especial prominence and substantially under-sampling Aboriginal cultural values and recreational fishing, which are known to be important to the region. Some social values, especially intangible values, may not be detectable at all in shared photos. Additionally, Flickr may fail to represent social values in areas with limited access or seasonal closures, given strong spatial biases in the sample, and was found to poorly sample the large marine domain of our study area. There is also considerable uncertainty around which stakeholders actually are represented by Flickr, and important stakeholder groups may be excluded from its sample.

PPGIS allows managers to design surveys around particular information needs (Heikinheimo et al., 2020), have greater understanding of the contributors (Levin et al., 2017), and may avoid access limitations. However, the quality of the data obtained can be influenced by the survey design, including the base maps provided, and the motivations and knowledgeability of the participants (Crouzat et al., 2022; van Riper et al., 2020). These concerns were supported by our results, which suggested that contributors (possibly those with only superficial knowledge of the Kimberley region) were unduly influenced by the protected area boundaries delineated on base maps and, consequently, valued ecosystem services in some locations where they did not use them. Such problems due to survey design or participant knowledgeability may be avoided in social media data (Dunkel, 2015; Ghermandi and Sinclair, 2019; Wang et al., 2022). PPGIS may be especially well-suited for gathering information about the social acceptability of activities across a region (Brown, 2017), directly contributing to participatory planning, improving community engagement and support for management actions. For example, the PPGIS dataset used in this study also requested participants to map their management preferences, identifying locations where particular activities should be increased or reduced for improved marine spatial planning to avoid conflict and ensure sustainability of social values in the Kimberley region (Strickland-Munro et al., 2016).

There is no single ‘right way’ to assess ecosystem services (Costanza et al., 2017; Harrison et al., 2018). Alternative crowdsourced means for sampling social values, such as Flickr photos and PPGIS surveys, provide differing perspectives and neither is a panacea. However, with thoughtful application and interpretation of results, they are valuable, efficient tools that can be used singly or in combination to inform spatial planning and management. We make the following recommendations for practical applications of crowdsourced social values data. Although our recommendations are primarily framed around the use of information shared to social media platforms such as Flickr, they are largely relevant to PPGIS data as well.

1. The spatial information about where ecosystem services are valued may be used with reasonable confidence, provided there is an adequate sample, especially if statistical modelling is used to spatially generalise from the point-based occurrences of social values mapped from geotagged social media posts and to account for gross spatial biases caused by patterns of accessibility. We found that, for the well-sampled terrestrial portion of our study area, the spatial predictions of suitable areas from our maxent models were reasonably robust between Flickr and PPGIS datasets for most of the social values. Although models based on the Flickr sample tended to identify a more constrained area as suitable for each social value, those areas were also generally identified to be suitable by models based on the PPGIS sample. However, absences are more uncertain than presences and, given the spatial limitations of social media samples, should not be interpreted as meaning an ecosystem service is not used or valued at a site.
2. Where possible, we recommend augmenting the information derived from social media posts with PPGIS or other traditional social science methods. Flickr and PPGIS approaches to crowdsource social values data are complementary; thus, we echo calls to use them synergistically (Depietri et al., 2021; Dunford et al., 2018; Fagerholm et al., 2021; Ghermandi and Sinclair, 2019; Heikinheimo et al., 2020; Komossa et al., 2020; Stahl Olafsson et al., 2022; Zhang et al., 2022). The potential benefits of doing so are four-fold: PPGIS surveys may fill gaps in the areas identified to be valued by social media samples by better capturing information about less popular sites (Depietri et al., 2021). Interviews and focus groups may enable the inclusion of important stakeholder groups, such as Aboriginal Traditional Owners (e.g., Moore et al., 2017), who are likely to be poorly represented by social media or online surveys. Photo-elicitation techniques may provide information about intangible values (e.g., Tonge et al., 2013) and how they are represented in photos, thereby improving the ability to detect these values in later content analyses of photos shared to social media. Finally, we advocate the use of social surveys to strengthen understanding of the socioecological drivers of social values. We found considerable uncertainty in the socioecological drivers identified by statistical models of social values, given the large differences in the variable usage of our maxent models constructed with Flickr vs. PPGIS samples. Thus, we recommend corroborating such inferences with surveys in which participants explain their motivations for using ecosystem services in the identified locations.
3. Finally, we recommend that investigators and managers actively engage with social media users to strengthen the contributed data and reduce uncertainties in their use. For example, managers could hold drives that solicit photos of particular social values or sites that are under-represented on social media to be shared and tagged accordingly (e.g., Nummi, 2018; and also advocated by Di Minin et al., 2015; Hausmann et al., 2018; Heikinheimo et al., 2017). Likewise, they could request that visitors share richer text with their photos to aid content analysis by providing greater information about the motivations for the photo and the experiences providing values, especially intangible values.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

Social values datasets were developed and evaluated in compliance with Murdoch University’s Human Research Ethics standards, Approval 2020/030 (Flickr data collection and analysis) and 2015/014 (PPGIS). PPGIS participants were active contributors and provided informed consent to participate in the research; Flickr users contributed passively to the research through public sharing of photos on Flickr’s platform, with consent to Flickr’s terms and conditions. The PPGIS dataset was funded by Kimberley Marine Research Program and Murdoch University, administered by the Western Australian Marine Science Association, but this research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. We thank Prof. Lynnath Beckley and Dr. Jennifer Munro for helpful feedback on previous versions of this work, and Dr. Michael Hughes for helpful discussions. We also thank the participants of the PPGIS project and contributors to the Flickr repository.

Appendix A-F. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ecoser.2023.101566.

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